

Zachary J. Levesque Ottawa, ON zlevesque93@gmail.com linkedin.com/in/zachary-levesque github.com/Zachary-Levesque yourwebsite.com		
TECHNICAL SKILLS Programming: Python, C, C++, C#, Java, Assembly, Verilog, TypeScript, JavaScript, HTML, MATLAB AI & ML: Neural Networks, PyTorch, TensorFlow, Qiskit, Training, Validation & Deployment, Quantum Computing Simulation: CAD, TCAD, FEM, Simulink, Multisim, Mathematical Modeling Circuits & IC Design: Digital + analog design, analysis & synthesis (Verilog, Synopsys), signal processing Hardware & Embedded Systems: Arduino, STM32, ESP, R-Pi, optical linecards, embedded systems, I2C/SPI/UART protocols Software & Tools: FastAPI, React.js, PostgreSQL, Docker, Git, Linux Professional: Data analysis, problem solving, leadership, bilingual (French, English)		
EDUCATION		
University of Ottawa <i>B.Sc./B.A.Sc. (Double Major) in Electrical Engineering and Physics</i>		Ottawa, ON Sep 2022 – Expected Apr 2027
- CGPA: 9.92/10 (4.0 GPA) 8× Dean's List (2022–2026); multiple Faculty of Engineering & Science Recognitions of Excellence - Awarded the International Experience Bursary to attend NCKU Digital IC Design & Synthesis (Taiwan, 2025): 98% best in class - ASCOT Recognition of Excellence Scholarship, Electro Federation Canada (2024) - Admission Scholarship, University of Ottawa (2022) – awarded for a 98% average (4.0 GPA)		
EXPERIENCE		
Ranovus Inc. <i>Systems Verification Engineer</i>		Kanata, ON May 2026 – Present
- Design and develop embedded firmware and board-level communication protocol stacks (I2C, SPI) for MCU-based hardware platforms - Perform PCB-level signal analysis and hardware debugging, backed by automated Python test scripts and validation infrastructure		
Seets Inc. <i>Co-Founder & CEO</i>		Ottawa, ON May 2026 – Present
- Co-founded and lead an IoT occupancy-tracking startup, owning product vision, system architecture, and full-stack engineering execution - Took the product from capstone concept to a live pilot deployment at the University of Ottawa Heart Institute		
APRIQuOt <i>Quantum AI Researcher (Intern)</i>		Ottawa, ON Sep 2025 – Apr 2026
- Applied machine learning to quantum state tomography, training neural networks to reconstruct multi-qubit quantum states from measurement data - Used Qiskit to simulate and study quantum chaos in complex multi-qubit systems, supporting ongoing quantum information research		
University of Ottawa <i>Teaching Assistant – ELG 3525, PHY 1722</i>		Ottawa, ON Sep 2025 – May 2026
- Led tutorials across two courses for both engineering (Signal & System Analysis) and physics (Physics II) undergraduates - Provided 1:1 and small-group academic support to students through the Physics Help Centre		
Huawei <i>Systems Engineer – Optical Communications (Intern)</i>		Ottawa, ON May 2025 – Aug 2025
- Conducted R&D on high-speed optical data transmission, designing communication links using EDFA, WSS, transceivers, and OSA hardware - Built Python automation for link control and data acquisition, applying statistical analysis to boost throughput and measurement turnaround		
Thin-Film Electronics Laboratory <i>Semiconductor Engineer (Intern)</i>		Ottawa, ON Jan 2025 – Apr 2025
- Developed TCAD physical simulation models to characterize non-linear charge injection behavior in n-type organic transistors - Built Python and MATLAB tools to model, analyze, and improve transistor performance and efficiency		
Hydro Ottawa <i>Data Engineering Intern</i>		Ottawa, ON May 2024 – Aug 2024
- Analyzed GIS data for the city's electrical distribution network to guide infrastructure planning and development - Built ML models to automate crew placement, material and tool allocation, project tracking, and budget forecasting		
PROJECTS		
Knovara <i>FastAPI, ChromaDB, OpenAI API, Next.js, TypeScript</i>		May 2026 – Present
Building an AI copilot that helps new engineers ramp up and become productive faster, piloting with Seets as first customer		
Mini Tensor Engine <i>C++, AVX2 SIMD, Python</i>		Jan 2026 – May 2026
Built a custom C++ engine from scratch that runs AI computations over 3× faster through low-level performance optimization		
Quantum Math Quest <i>Python</i>		Nov 2025
Built an adaptive learning game that adjusts difficulty in real time, blending game design with quantum-inspired concepts		
Adaptive Market Regime Framework <i>Python, HMMs, RL</i>		Aug 2025 – Dec 2025
Built a machine learning system that detects shifts in market conditions and adjusts strategy automatically, beating the S&P 500 benchmark		
Sports Strategy Engine <i>Python, FastAPI, React, SQLite</i>		May 2025 – Sep 2025
Built a golf strategy tool that simulates thousands of outcomes to recommend the optimal shot under real-world uncertainty		
Real-Time Trading System <i>FastAPI, React, PostgreSQL, Redis</i>		Jan 2025 – May 2025
Built a full-stack platform that analyzes live market data and generates clear, explainable buy/hold/sell recommendations		
AI Games & Logic Solvers <i>Python, PyTorch</i>		Sep 2024 – Dec 2024
Built a collection of AI agents that master classic games and solve complex logic puzzles using advanced decision-making algorithms		
AI Agent/Workflows <i>Python</i>		Apr 2024 – Jun 2024
Built a suite of AI-powered tools that automate scheduling, lead generation, and research, saving hours of manual work		
ParaPicture Solutions <i>Hardware Prototyping</i>		Sep 2023 – Dec 2023
Engineered an assistive robotic arm that lets individuals with limited mobility independently take photographs		
1Chat4You <i>TypeScript</i>		Jan 2023 – Apr 2023
Built a communication platform that makes coordinating between landlords and tenants seamless and secure		
CircuitFlow <i>TypeScript, Circuit Design & Analysis</i>		Sep 2022 – Dec 2022
Built a real-time circuit simulator that lets users design and analyze electrical circuits interactively		